MedVAE: Adaptation, Fine-tuning and Analysis for Medical Imaging and Coronary Segmentation

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***Abstract*—MedVAE is a generic medical autoencoder trained on chest X-rays and mammograms. We study how far it can be trusted, adapted and reused on an out-of-distribution modality, coronary angiography (ARCADE dataset), along four axes, with three seeds for every trained model and control baselines. (i) Robustness: MedVAE reduces Poisson noise, but 3.4 to 6.3 dB less than a simple 5×5 filter, and restores neither JPEG artefacts nor blur. (ii) Image-quality conditioning: fine-tuning MedVAE on ARCADE gains 7.4 dB, while injecting a quality score through FiLM adds at most a few hundredths of a decibel, and the network barely uses the score. (iii) JEPA adaptation: replacing the BioMedCLIP stage with a self-supervised JEPA objective lowers both the reconstruction and the linear separability of the vessels. (iv) Segmentation: segmenting MedVAE reconstructions matches the original images (artery Dice 0.424 against 0.428), unlike a naive downsampling of the same rate (0.389); its single-channel latent reaches 96 % of the full-image Dice, about as much as an image downsampled to the same size. MedVAE's compression is thus anatomically faithful in pixel space, but neither quality conditioning nor JEPA adaptation improves it in the forms we tested.**

***Index Terms*—medical image compression, variational autoencoder, coronary angiography, robustness, FiLM conditioning, JEPA, segmentation**

Code available at: https://github.com/tnadiedjoa/MEDVAE-X

I. Introduction

Deep learning on medical images is constrained by the high resolution required to preserve diagnostic detail, which inflates the computational and memory cost of every model. Large-scale variational autoencoders (VAE) offer a way out by compressing images into compact latent representations. We focus on MedVAE [1], a representative example: trained on more than one million medical images, it learns a generic, reusable compression of medical images — in the variant we use, reducing a 512×512 image by a factor of 4 per side (sixteen times fewer values) into a single-channel 128×128 latent (the FiLM study of Section III uses the 4-channel variant) — while seeking to preserve clinically relevant structures, and is itself built in two stages: a first stage dedicated to faithful reconstruction, and a second stage meant to enrich the latent space with semantic knowledge. Such generalist models raise a natural question, which we take as our starting point: to what extent does a generic medical compressor remain trustworthy and useful once confronted with a specific, demanding clinical domain?

We study this question on coronary angiography, using the public ARCADE dataset [2], which provides $1\,500$ annotated X-ray angiographies (512×512, grayscale; $1\,000$ training, 200 validation and 300 test images) together with COCO segmentation annotations of the coronary tree. We deliberately chose an unfavourable setting for a generic compressor: the structures of interest are thin, tubular, and low-contrast, the acquisitions suffer from quantum noise, compression, and motion artefacts, and the relevant signal occupies only a tiny fraction of the image. This domain therefore stresses every property we would expect from a medical foundation model: robustness to degraded inputs, a semantically structured latent space, and the preservation of fine anatomical information.

Starting from MedVAE and ARCADE, we identify and investigate four complementary axes. **(i) Robustness to input degradations** (Section II): we measure whether MedVAE brings a degraded input (Poisson noise, JPEG compression, blur) closer to the clean image, showing that it reduces Poisson noise, though much less than a simple 5×5 filter, and restores neither JPEG artefacts nor blur. **(ii) Image-quality inductive bias** (Section III): we ask whether explicitly conditioning the autoencoder on an estimated quality score, injected through FiLM modulation, lets it adapt to the degradation level actually present at its input. **(iii) Semantic structuring of the latent space** (Section IV): we replace MedVAE's stage-2 BioMedCLIP alignment with a self-supervised JEPA objective trained directly on coronary angiographies, removing any dependency on an external model. **(iv) Latent representation for segmentation** (Section V): we test whether the compressed representation preserves enough anatomical information to support fine multi-class coronary segmentation, and at which point of the pipeline the compression should be integrated.

We each took on one of these axes — Elias Corlou the robustness analysis, Théophile Nadiedjoa the image-quality conditioning, Yanic Rothlingshofer the JEPA adaptation, and Théo Palagi the segmentation study — and we present our four contributions in the corresponding sections before bringing them together in a common discussion of perspectives and a shared conclusion.

II. Robustness of MedVAE to Input Degradations

MedVAE was evaluated, in its original paper [1], on clean inputs — by PSNR and MS-SSIM as well as by expert reader studies. One question remains open for real-world deployment: *what does MedVAE do with an input image that is not clean, but degraded?* In coronary angiography, this question is far from theoretical: low-dose acquisition induces significant quantum noise, and clinical images undergo compression, motion blur, and various artefacts [3].

We evaluate this on the ARCADE dataset [2]. The framework is entirely *full-reference*: for each image we simultaneously have the clean image, a controlled degraded version, and the reconstruction produced by MedVAE from the degraded input. The research question is: **does passing a degraded image through MedVAE bring it closer to, or further from, the clean image?** We show that the answer depends on the *nature* of the degradation: MedVAE reduces Poisson noise, though much less than a simple 5×5 filter, and restores neither JPEG artefacts nor blur.

Note on this revised version. An earlier version of this study measured the fidelity of the reconstruction to the *degraded* input, with a per-image min-max normalisation that placed the clean and degraded images on different intensity scales. Both choices biased the conclusions (Section II-A); the protocol and all results below were redone accordingly.

A. Evaluation Framework

1) Why No-Reference Metrics Were Discarded: A first, natural approach consists of quantifying the quality of the input image by a *no-reference* metric, then correlating it with reconstruction quality. We explored this route with ARNIQA [4], as well as with a custom score aggregating sharpness indices. Two limitations disqualified it. On one hand, these metrics are calibrated on natural photographs and transfer poorly to X-ray images: on the raw images of the dataset, the correlation with the reconstruction PSNR is weak and of the wrong sign for ARNIQA (Pearson $r = -0.37$) and nearly zero for our custom score ($r \approx -0.01$). On the other hand, under controlled degradation, these scores are dominated by blur and do not provide a monotone quality axis [4]. Our protocol being full-reference, we instead *measure* input quality by comparison to the clean image.

2) Measuring Restoration, Not Fidelity to the Input: Three PSNR values can be computed for each degraded image:

- $\mathrm{PSNR}(\text{clean}, \text{degraded})$: how far the input is from the clean image (degradation amplitude);
- $\mathrm{PSNR}(\text{clean}, \text{reconstructed})$: how far MedVAE's output is from the clean image;
- $\mathrm{PSNR}(\text{degraded}, \text{reconstructed})$: how faithfully MedVAE reproduces the input it was given.

The third quantity, used in the earlier version of this study, does not measure robustness: a blurred image, stripped of its high frequencies, is simply easier to reproduce, so this fidelity rises with blur even if the output moves away from the clean image. The quantity that answers our question is the restoration gain

$$\Delta = \mathrm{PSNR}(\text{clean}, \text{reconstructed}) - \mathrm{PSNR}(\text{clean}, \text{degraded}), \tag{1}$$

positive when MedVAE's output is closer to the clean image than its input was. All images are compared on the same fixed intensity scale ($[0, 1]$, pixel value$/255$).

3) Vessel-Masked PSNR: A PSNR computed over the full image mostly reflects the fidelity of the background, which occupies the vast majority of the surface. We therefore also restrict the computation to vessel pixels, obtained by rasterising ARCADE's polygonal annotations (COCO format) into a binary mask, which defines a masked PSNR denoted mPSNR. Every quantity above is computed both on the full image and in this masked version (details in Appendix B).

B. Method

1) Protocol: We use the `medvae_4_1_2d` autoencoder (X-ray modality) on 100 images of ARCADE's official segmentation validation split (`seg_val`). Each image, scaled to $[-1, 1]$ as MedVAE expects, is encoded and decoded *deterministically* (mean of the approximate posterior, instead of a random sample). For each degradation type, 10 levels of increasing severity are applied, the noise being drawn with a generator seeded per image and per level; the clean image is also reconstructed as a reference. Metrics are averaged over images, with 95% confidence intervals obtained by bootstrap over images. The code is in `medvae_eval/robustness/`.

C. Degradation Types

Three degradations are studied, each swept independently, over the same ranges as in the earlier version of the study:

Poisson noise. More realistic than Gaussian noise for X-rays, where quantum noise (*shot noise*) dominates at low dose [3]. Intensity is controlled by a decreasing scale factor (from 1 to 0.05): as the scale decreases, fewer photons are simulated and the noise becomes more pronounced.

JPEG compression. Quality factor decreasing from 95 to 5, successively introducing quantisation of DCT coefficients and then 8×8 block artefacts.

Gaussian blur. Kernel size increasing from 1 (clean image) to 31 pixels.

D. Results and Interpretation

MedVAE reconstructs these clean images at 33.1 dB, both on the full image and on vessels (34.8 dB on the 300 test images used in Section IV): this intrinsic reconstruction error sets a ceiling that no restoration can exceed. As a reference, the same noisy images were also passed through two simple filters, a 5×5 Gaussian filter and a 5×5 median filter. Fig. 2 shows the three PSNR curves and the restoration gain Δ on vessel pixels, and Table I summarises Δ over the sweep.

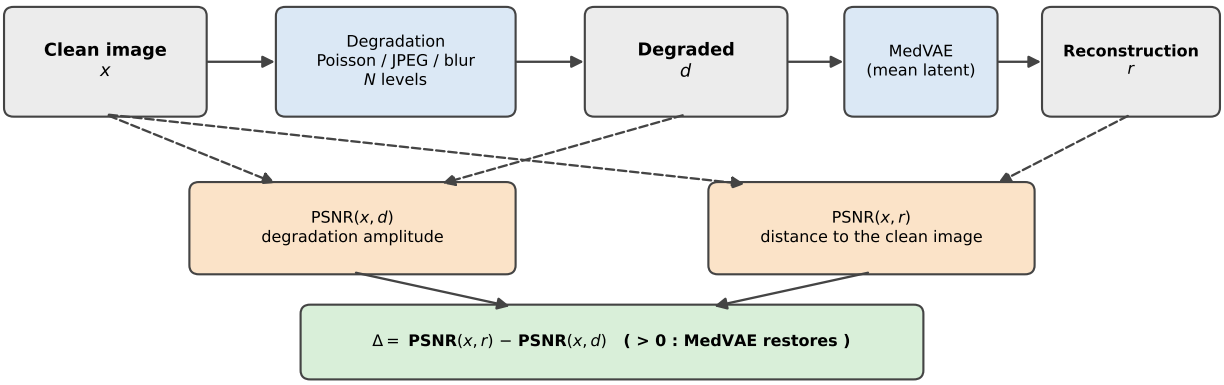


Fig. 1: Evaluation pipeline. A batch of clean images is degraded in a controlled manner over N levels, then reconstructed by MedVAE. The reconstruction is compared to the *clean* image, and the restoration gain Δ (Eq. 1) compares this distance to that of the degraded input. A separate sweep is conducted per degradation type (Poisson, JPEG, blur).

TABLE I: Restoration gain Δ (dB, Eq. 1) from the weakest to the strongest degradation level, and share of images restored ($\Delta > 0$, full image), for MedVAE and, on Poisson noise, for two simple filters. 95% confidence intervals are at most 0.53 dB wide for MedVAE.

Degradation	Δ full image	Δ vessels	Restored
Poisson noise	$+1.2 \to +6.6$	$+0.9 \to +6.8$	99–100%
Gaussian filter 5×5	$+6.3 \to +10.3$	$+7.1 \to +10.9$	100%
Median filter 5×5	$+7.2 \to +10.0$	$+5.7 \to +10.3$	100%
JPEG ($95 \to 5$)	$-11.7 \to -0.5$	$-11.3 \to -0.1$	0–9%
Blur (kernel $5 \to 31$)	$-2.0 \to +0.1$	$-2.8 \to -0.2$	3–82%

1) Poisson Noise: Reduced, Less Than by a Simple Filter: Poisson noise adds decorrelated high frequencies, precisely what the latent bottleneck cannot represent, so MedVAE suppresses it [5]. The reconstruction is closer to the clean image than the noisy input for 99 to 100% of the images at every level, and the gain grows with the noise, from $+1.2$ dB at the weakest level to $+6.6$ dB ($+6.8$ dB on vessels) at the strongest. This noise reduction is, however, modest: a 5×5 Gaussian filter gains $+6.3$ to $+10.3$ dB on the same images and a 5×5 median filter $+7.2$ to $+10.0$ dB, 3.4 to 6.3 dB more than MedVAE at every level. MedVAE's gain is bounded by its own reconstruction error: even for mild noise, the output (28 dB) stays well below the reconstruction of the clean image (33 dB).

2) JPEG Compression: No Restoration: JPEG artefacts are small in PSNR terms (the input is at 44.7 dB of the clean image at quality 95), far below MedVAE's own reconstruction error. Passing the image through MedVAE therefore always moves it away from the clean image, by 11.7 dB at quality 95 and 0.5 dB at quality 5. The earlier observation of a "two-regime" response concerned the fidelity to the degraded input and does not translate into any restoration.

3) Gaussian Blur: No Restoration: Blur removes high frequencies that the latent cannot recover. The reconstruction simply follows the blurred input: Δ is negative for mild blur (the reconstruction error adds to the blur) and close to zero for strong blur on the full image, while it remains slightly negative on vessels. The earlier finding that "blurred inputs reconstruct better" is visible in the orange curve of Fig. 2 (fidelity to the blurred input rising from 33.1 to 47.7 dB on vessels): it measures how easy a blurred image is to reproduce, not how well MedVAE recovers the clean one, which is the opposite trend (blue curve).

4) Vessel Visibility: Contrast-to-Noise Ratio: A 5×5 filter blurs thin vessels, so its PSNR advantage might not mean that vessels remain more visible. We therefore measure the contrast-to-noise ratio of the vessels, a standard quality measure in angiography: $\mathrm{CNR} = (\mu_{\mathrm{bg}} - \mu_{\mathrm{vessel}})/\sigma_{\mathrm{bg}}$, where μ_{vessel} is the mean intensity on the vessel centrelines (skeleton of the annotation mask) and μ_{bg}, σ_{bg} the mean and standard deviation of a local background ring, 3 to 13 pixels around the mask. We separate thin vessels (annotated width at most 6 pixels, the lowest quartile) from thick ones (at least 12 pixels, the highest quartile), on the same images, degradations and noise as above (Table II).

TABLE II: Vessel CNR (mean over the 100 images) for thin and thick vessels: degraded input, MedVAE, and 5×5 Gaussian and median filters. Clean image: 0.88 (thin) and 1.91 (thick); MedVAE on the clean image: 0.85 and 1.90.

	Thin vessels				Thick vessels			
	Input	MedVAE	Gauss.	Med.	Input	MedVAE	Gauss.	Med.
Poisson, level 0	0.72	0.73	**0.81**	0.76	1.54	1.70	1.97	**2.00**
Poisson, level 6	0.58	0.64	**0.77**	0.72	1.25	1.53	1.90	**1.92**
Poisson, level 9	0.30	0.45	**0.62**	0.55	0.62	1.16	**1.49**	1.48
JPEG, level 6	**0.88**	0.86	0.83	0.79	1.94	1.90	2.01	**2.02**
Blur, level 6	0.50	0.50	0.49	0.48	1.92	1.92	1.90	1.89

The CNR confirms the PSNR ranking, thin vessels included. Under Poisson noise, MedVAE makes vessels more visible than the noisy input, clearly for thick vessels ($+0.16$ to $+0.54$) but barely for thin ones ($+0.01$ to $+0.15$), and the Gaussian

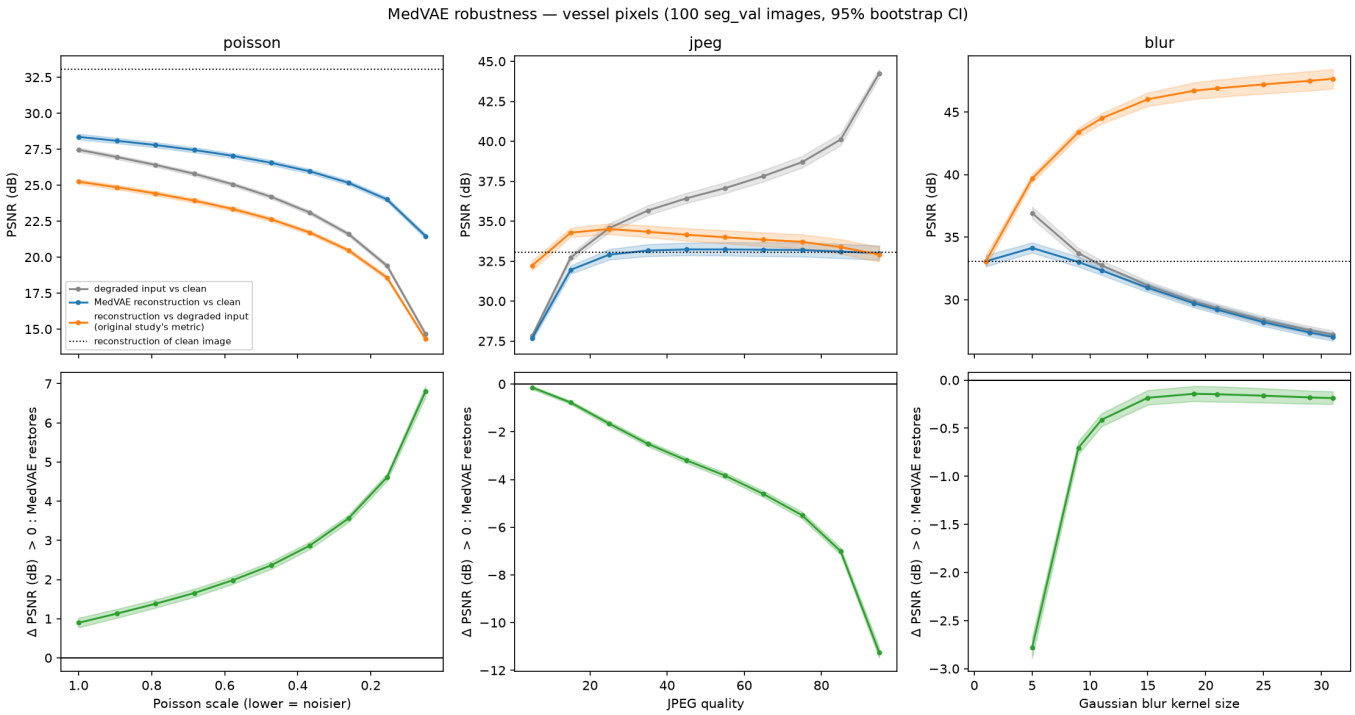


Fig. 2: Robustness of MedVAE on vessel pixels (100 `seg_val` images, 95% bootstrap confidence intervals). Top: PSNR of the degraded input (grey) and of MedVAE's reconstruction (blue) with respect to the clean image; in orange, the fidelity of the reconstruction to the degraded input, the measure used in the earlier version of this study. Bottom: restoration gain Δ; positive values mean that MedVAE brings the image closer to the clean one.

filter does better at every level, by 0.08 to 0.18 on thin vessels and 0.27 to 0.39 on thick ones (95% bootstrap confidence intervals excluding zero). Both keep the contrast of thick vessels, but the filter removes more noise; on thin vessels, MedVAE loses up to a third of the contrast at the strongest noise, where the filter keeps about 90%. Without noise to remove (clean image, JPEG, blur), MedVAE leaves the CNR almost unchanged, while the filters, which only blur, lower that of thin vessels. Two caveats: on clean images, the background standard deviation mostly reflects anatomical texture rather than quantum noise, and the CNR rewards smoothing (the Gaussian filter exceeds the clean image on thick vessels); a task-based measure addresses both (Section II-E).

E. Discussion and Perspectives

MedVAE acts as a low-pass autoencoder: it attenuates stochastic high frequencies (Poisson noise), though less than a simple 5×5 filter, it cannot recover information removed by blur, and its own reconstruction error exceeds the mild artefacts of JPEG compression. Beyond the degradation itself, the 33 dB reconstruction ceiling is the main limit of any restoration through MedVAE on this out-of-distribution modality.

These characteristics call for several clinical evaluation perspectives. First, it will be necessary to quantify the impact of this low-pass dynamic on stenosis detection, the central task of the ARCADE dataset, as these pathologies constitute fine structures particularly sensitive to smoothing; measuring segmentation quality on degraded images shows that MedVAE's denoising does not make segmentation more robust (Section V). Furthermore, evaluating robustness to alterations of the grey-level dynamic range (gamma variations, underexposure) is a natural next step. The protocol would also benefit from integrating motion blur, induced by cardiac and respiratory dynamics, whose spectral signature is far more complex than the isotropic blur modelled here. Finally, PSNR, vessel PSNR and CNR remain proxies for what matters clinically: whether a vessel can still be seen. Task-based image quality assessment answers this question directly [6]. A model observer, such as the channelized Hotelling observer, decides whether a vessel is present in an image patch, and its detectability index d', the separation between its responses with and without the vessel, accounts for both the noise, including its texture, and the resolution, which the CNR ignores; it would also penalise vessel-like structures hallucinated by the autoencoder. Inserting synthetic vessels of controlled width and contrast into real ARCADE backgrounds, degrading them at several dose levels and comparing d' with and without MedVAE would tell whether MedVAE preserves the detectability of thin vessels better than a simple filter, and would express its effect as an equivalent dose.

III. Image-quality inductive bias for MedVAE

A. Motivation

The previous section showed that MedVAE removes Poisson noise but restores neither JPEG artefacts nor blur, and that it has no explicit way of knowing *how much* the image it is given is degraded: it applies the same transformation to a clean or a heavily noisy angiography. Yet this information is available *a priori* and can be estimated by no-reference quality metrics (NR-IQA). We hypothesise that explicitly conditioning the autoencoder on a quality score $c \in [0,1]$ improves its reconstruction, by letting it adapt its behaviour to the degradation level actually present at the input. Three questions follow: how to compute c objectively and reproducibly in the absence of human annotation on ARCADE; how to inject it into MedVAE without retraining the model from scratch; and whether any observed gain is attributable to the informative content of c or to a mere additional-capacity effect. This last question is far from trivial: any conditioning mechanism adds parameters and changes the training dynamics, so an apparent gain could simply reflect increased capacity rather than a genuine exploitation of the quality signal, hence the need for a dedicated ablation (section III-F). For the first question, three independent methods for computing c (approaches A, B, C) are compared, each with three training seeds, which makes it possible to distinguish an effect specific to a given method of computing c from a more general effect of the FiLM conditioning itself.

B. Construction of the quality score

On the segmentation subset of ARCADE (1000 train / 200 validation images, 512×512, grayscale), each image is split into a 4×4 grid of patches, a local granularity motivated by the spatially non-uniform nature of artefacts in cardiac imaging (motion, noise) [7], normalised by *Mean Subtracted Contrast Normalization* ($\hat{I} = (I-\mu)/(\sigma+C)$, 7×7 window), then characterised by 8 metrics per patch: Laplacian variance, RMS contrast, Shannon entropy, and the Tenengrad gradient broken down into 4 directions, a directional analysis suited to the tubular structure of vessels [8] (which allows detection of anisotropic cardiac motion blur). These per-patch metrics are then aggregated over the patches of each image (mean, minimum and similar statistics). A pre-trained ResNet-18 (frozen weights) extracts in parallel a 512-dimensional feature vector, used only by the learned approach: the two branches of this feature pipeline (Figure 3) never converge directly, they simply feed different score-computation methods, described below.

These features are then merged into a single score c, with 1 corresponding to a clinically usable image, using three independent methods (Figure 4), all fitted exclusively on the TRAIN subset. Approach B applies a PCA (after standardisation) on 6 selected features (four of the aggregated metrics — mean Laplacian variance, minimum RMS contrast, inverse entropy and mean Tenengrad — plus a spatial-homogeneity and a directional-balance feature); the first axis explains

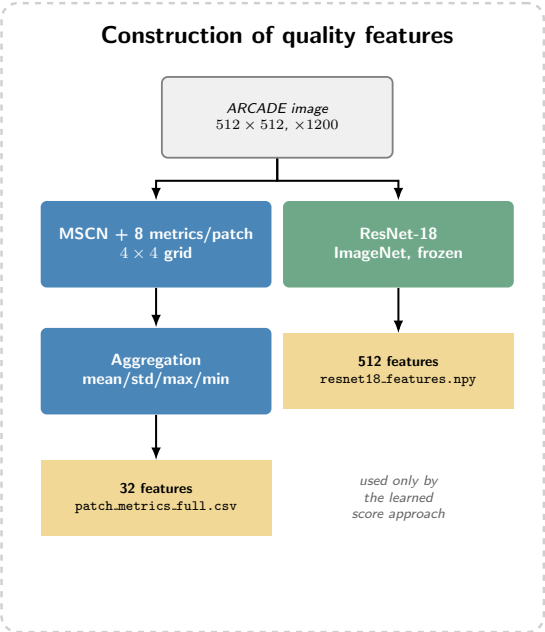


Fig. 3: Construction of quality features: per-patch IQA metrics and ResNet-18 features.

56.1 % of the variance, and its sign is corrected by correlation with Tenengrad ($r = 0.952$). Approach A linearly weights the same 6 features by the absolute normalised loadings of this same PC1. Approach C, lacking human annotation on ARCADE, learns a score on a *synthetic proxy*, an indirect supervision strategy already used to calibrate NR-IQA scores on biomedical images without perceptual ground truth [9]: each TRAIN image is artificially degraded (noise, blur, JPEG) with a target $y = 1 -$ known severity, and a lightweight MLP ($512 \to 128 \to 32 \to 1$) is trained by MSE on the ResNet-18 features of this synthetic set (validation MSE 1.58×10^{-3}), then applied to the real images.

A and B, built on the same metrics, are strongly correlated ($r = 0.884$); C, in contrast, is nearly independent of both ($r(A,C) = 0.003$, $r(B,C) = 0.015$, Table III) and its distribution is much more tightly concentrated near 1 (mean 0.890) than A (0.448) or B (0.483): the MLP learned on deep features captures a notion of quality that differs from a combination of classical metrics. Figure 5 illustrates this progression on real images sorted by increasing score: the least sharp and noisiest angiographies (left) progressively give way to contrasted acquisitions where the vascular tree is clearly defined (right), which qualitatively suggests that the score tracks visible sharpness and contrast.

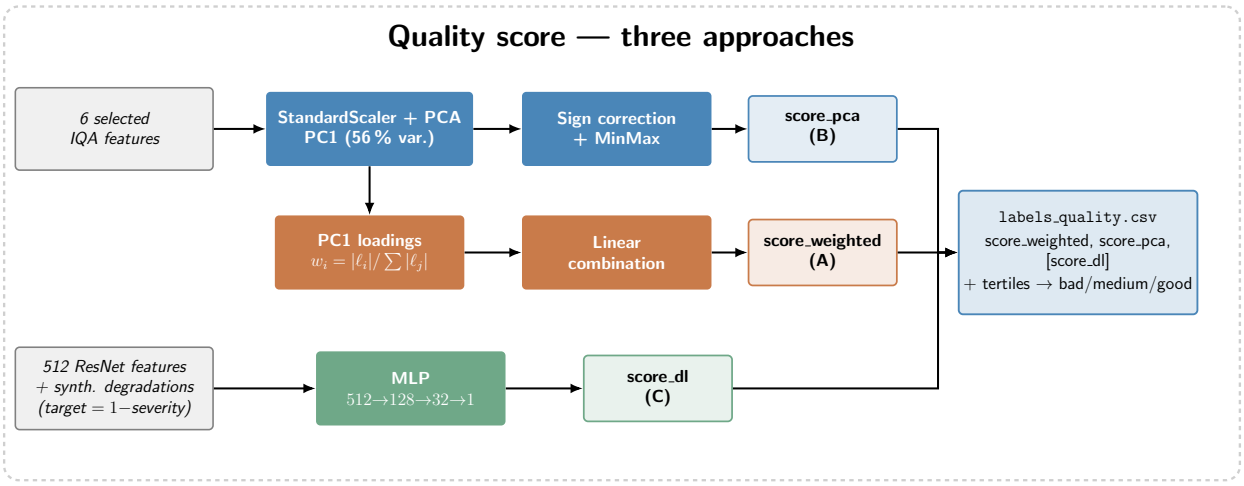


Fig. 4: The three approaches for computing the quality score c.

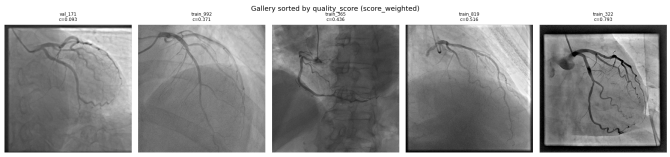


Fig. 5: ARCADE images sorted by increasing quality score (approach A, weighted): from least sharp (left) to most contrasted (right).

	A (weighted)	B (PCA)	C (learned)
A (weighted)	1	0.884	0.003
B (PCA)		1	0.015
C (learned)			1

TABLE III: Pearson correlation between the three scores (1200 images).

C. FiLM conditioning

To condition MedVAE [1] on c without modifying its input topology, we use *Feature-wise Linear Modulation* (FiLM [10]) rather than channel concatenation. We work with the `vae_4x_4c_2D` variant of MedVAE (`AutoencoderKL` [11], factor 4 per side, 4 latent channels, 55.3M parameters). A shared MLP encodes c into a 128-dimensional embedding, and 19 linear heads (one per `ResnetBlock` of the encoder, bottleneck, and decoder) produce affine parameters applied through a hook on each block, $h \leftarrow (1 + \gamma_i(c)) \cdot h + \beta_i(c)$. The heads are initialised at zero, so the conditioned model is identical to the original model before any training: any difference observed afterwards stems from what the network has learned, not from a random perturbation of the pre-trained weights. FiLM also leaves the encoder's topology intact and concentrates the conditioning on clearly identifiable parameters. This mechanism adds 1.70M parameters (3.1 %).

D. Protocol

Note on this revised version. An earlier version of this study trained and evaluated all models at 64×64, a resolution at which the pre-trained MedVAE only reaches 20.7 dB on our test images (against 34.8 dB at 256×256) and vessels shrink to about one pixel. Its loss added the KL divergence *summed* over the latent to a *per-pixel mean* squared error, so that the KL term dominated the objective (at 256×256 it outweighs the reconstruction error by a factor of about 700 and makes the validation error rise during training); the conditioned model and the baseline followed different training schedules; and every comparison relied on a single training run per model. The protocol and all results below were redone accordingly.

Data. The 1000 images of the ARCADE training split are divided once and for all into 900 training and 100 validation images; the 200 images of the official validation split serve as the *test set*, never seen during training or model selection. Images are resized to 256×256.

Training. The baseline (MedVAE without conditioning) and the FiLM model are trained in exactly the same way: all weights, 3000 steps, AdamW (learning rate 10^{-5}, cosine decay to 10^{-6}), batch size 4, loss $=$ MSE $+ 10^{-6}\,\mathrm{KL}/N_{\text{pixels}}$ (the KL is normalised by the number of pixels, following the MedVAE and latent-diffusion convention). The checkpoint with the lowest validation reconstruction error is kept. Each model is trained with 3 seeds.

Evaluation. On the test set, we compute the PSNR, the PSNR restricted to vessel pixels (mPSNR, Section II-A3), SSIM and HaarPSI, the latter being recommended for medical images [12]. To test whether the FiLM model uses the *content* of c, each FiLM model is evaluated three times (Figure 6): with the true score of each image, with the score of another test image (a fixed random permutation of the scores), and with a constant score (the training mean). If the network exploits the quality information, the true score must beat both controls. Unlike an ablation that trains a second model with a constant

score, these controls are computed on the *same* trained model, so their differences are free of the variability between training runs.

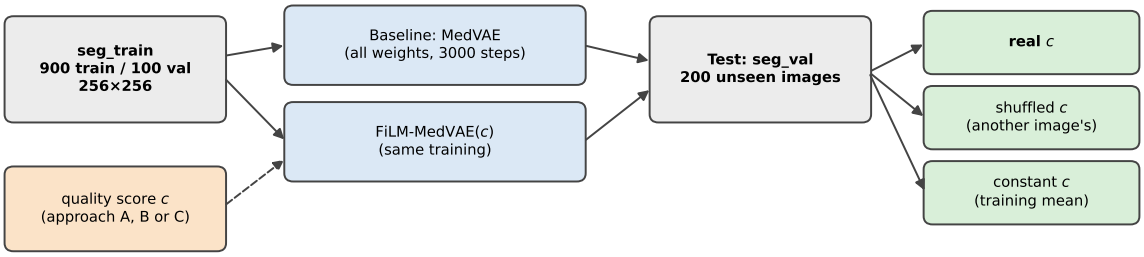


Fig. 6: Revised protocol: identical training for the baseline and the FiLM model, evaluation on unseen images, and two controls on c computed on the same FiLM model.

E. Results

Model	PSNR	mPSNR	SSIM	HaarPSI
MedVAE pre-trained	34.81	35.57	0.949	0.939
Baseline (fine-tuned)	42.26 ± 0.01	40.89 ± 0.01	0.970	0.983
FiLM, true c	42.29 ± 0.02	40.92 ± 0.01	0.970	0.983
FiLM, shuffled c	42.29 ± 0.02	40.91 ± 0.01	0.970	0.983
FiLM, constant c	42.30 ± 0.02	40.93 ± 0.01	0.970	0.984

TABLE IV: Reconstruction of the 200 test images (256×256, PSNR in dB), approach C for c; mean ± standard deviation over 3 seeds (SSIM and HaarPSI vary by less than 10^{-3}).

Table IV and Figure 7 give the results. Fine-tuning MedVAE on ARCADE is by far the largest effect: $+7.4$ dB of PSNR ($95\,\%$ CI over images: $[7.1, 7.9]$) and $+5.3$ dB on vessels relative to the pre-trained model, with the same gain for the three seeds. On top of this, FiLM conditioning changes almost nothing: $+0.03$ dB relative to the baseline (per seed $+0.055$, $+0.004$ and $+0.019$; $95\,\%$ CI over the seeds $[-0.04, +0.09]$), far below any perceptible difference. The earlier conclusion that the conditioned model reconstructs markedly worse than the baseline does not hold once both models are trained in the same way.

F. Is the quality signal used?

The controls answer this question directly (Table V). With score C, replacing the true score of each image by that of another image changes the PSNR by $+0.002$ dB ($95\,\%$ CI over images $[-0.007, 0.010]$), and a constant score does even slightly *better* than the true one (-0.013 dB). This was expected by construction: score C barely varies on the training images (mean 0.92, $79\,\%$ above 0.9, 5–$95\,\%$ range $[0.70, 1.00]$), whereas it is lower and more spread on the test images (mean 0.73, $26\,\%$ below 0.5). The model almost never saw a low score, and over the range it did see, the FiLM modulation γ varies by at most 0.02.

Scores A and B, whose distributions are similar on training and test images, were therefore studied with three seeds as well. With both, the true score beats the two controls for every seed: the network does use the quality information. The effect is, however, tiny: $+0.007$ dB against a shuffled score with A and $+0.033$ dB with B, the largest of all. Over the baseline, FiLM gains $+0.04$ dB with A and $+0.05$ dB with B, of which the quality information explains at most 0.02 to 0.03 dB; the rest comes from the added per-channel affine parameters. A last run with a learning rate 100 times higher for the FiLM layers (score C, one seed) confirms this reading: it gains 0.24 dB over the baseline, but a constant score does exactly as well as the true one.

PSNR (dB)	FiLM − baseline	true − shuffled c	true − constant c
Score C (3 seeds)	$+0.026$ $[-0.04, +0.09]$	$+0.002$	-0.013^*
Score A (3 seeds)	$+0.038$ $[-0.01, +0.09]$	$+0.007^*$	$+0.005^*$
Score B (3 seeds)	$+0.050$ $[-0.003, +0.102]$	$+0.033^*$	$+0.020^*$
Score C, FiLM lr ×100 (1 seed)	$+0.239$	$+0.011$	-0.001

TABLE V: Effect of FiLM and use of the quality score on the 200 test images. FiLM − baseline compares separately trained models: mean over seeds and $95\,\%$ CI over the seeds. The controls compare variants of the same model: * marks a $95\,\%$ bootstrap CI over images excluding zero (same sign for every seed).

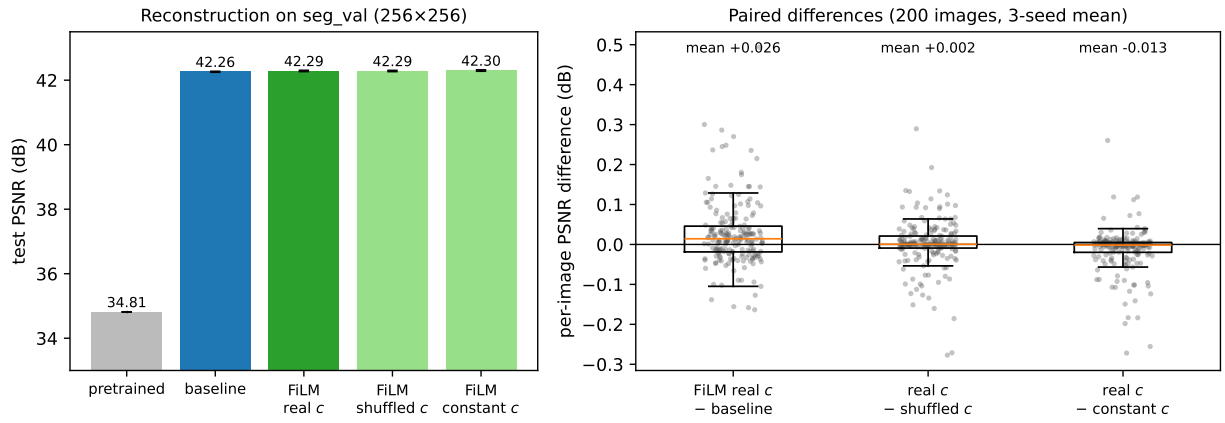


Fig. 7: Left: test PSNR per model (mean ± standard deviation over 3 seeds). Right: per-image paired PSNR differences, averaged over the 3 seeds; the dots are the 200 test images.

G. Discussion

Measured with identical training, an independent test set and controls computed on the same model, conditioning MedVAE on an image-quality score brings no useful gain in reconstruction: the learned score C, nearly constant on the training images, is not used, and scores A and B are used, but their effect stays below 0.04 dB. This revises both conclusions of the earlier version of this study (a conditioned model markedly worse than the baseline, and score C as the only one genuinely exploited): both were obtained under the conditions listed in Section III-D, and neither survives the revised protocol.

A simple reason limits what the score can bring. All three scores are computed *from the image itself* (quality metrics for A and B, ResNet-18 features for C): for an autoencoder whose target is its own input, c contains no information that the encoder does not already see. At most it summarises a global property of the image (overall sharpness or contrast) that a purely convolutional network only perceives locally, which may explain the small but consistent use of scores A and B. A quality signal could only bring new information in a setting where it is not a function of the input, for instance a restoration task (degraded input, clean target) conditioned on acquisition parameters known independently of the image, such as the dose. Two further limitations remain: a single training budget was explored (the validation error was still decreasing at the last of the 3000 steps for most runs), and the absence of human quality annotation on ARCADE means that the three scores could not be validated against human quality labels; A and B agree ($r = 0.884$, they share the same features), while C is unrelated to them ($r \approx 0$).

IV. JEPA Adaptation of MedVAE Stage 2

As introduced above, MedVAE [1] is trained in two stages, the second of which enriches the latent space with semantic knowledge — the stage we revisit here. Medical imaging raises specific challenges — monochrome images, scarce annotations, heterogeneous modalities — that make learning robust representations particularly difficult.

A. Motivations

In its original formulation, MedVAE stage 2 aligns the latent space with BioMedCLIP representations, a foundation model pre-trained on paired medical image–text corpora. In MedVAE, this stage freezes the encoder and decoder and trains only light projection layers on top of the latent, with an L_2 loss between their output and the BioMedCLIP image embedding of the same image; neither BioMedCLIP nor text is needed at inference. This approach has two limitations. First, training depends on an external model. Second, and more fundamentally, the alignment inherits BioMedCLIP's biases and coverage gaps: coronary angiography represents a narrow and specialised sub-domain in large-scale medical vision–language pretraining, so the transmitted semantic signal may be weak or misleading. Aligning to a model that has never seen enough coronary images is unlikely to produce representations that are discriminative for vascular anatomy.

JEPA (*Joint-Embedding Predictive Architecture*) architectures [13] offer a self-supervised alternative that removes the dependency on any external model. Rather than reconstructing masked pixels — which can waste capacity on low-level texture — JEPA predicts in latent space the representations of masked regions. This forces the encoder to model the contextual relationships between visible patches and occluded areas, learning representations that are inherently structured and high-level. The approach has shown strong results across medical imaging modalities: US-JEPA [14] for ultrasound, EchoJEPA [15] for echocardiography, Brain-JEPA [16] for fMRI, and Seg-JEPA [17] for segmentation.

We propose replacing stage 2's BioMedCLIP alignment with JEPA training directly on coronary angiography images, with no dependency on an external model. Following MedVAE's two-stage recipe on ARCADE, stage 1 trains an autoencoder of the MedVAE family from scratch with reconstruction, perceptual and adversarial losses (15 epochs on the 1 200 images of ARCADE's training and validation splits, 1 080 for training and 120 for model selection; the test images are never seen). This autoencoder is smaller than the released `medvae_4_1_2d` (64 base channels, four resolution levels, 21M parameters) and compresses by a factor of 8 per side, into a $64 \times 64 \times 1$ latent. Stage 2 starts from the stage-1 weights and keeps the decoder frozen throughout. Unlike MedVAE's stage 2, which trains only projection layers, it trains the encoder itself: our experiment therefore tests an *adaptation of the encoder* by JEPA, and the reconstruction can degrade by construction, since the frozen decoder was trained on the stage-1 encoder.

B. Method

a) Original stage 2 — BioMedCLIP alignment.: In the original MedVAE formulation (Fig. 8), stage 2 freezes the encoder and decoder and trains lightweight projection layers on top of the latent so that their output matches the BioMedCLIP embedding of the same image. This step aims to semantically enrich the latent space by transmitting the knowledge of a pre-trained medical foundation model.

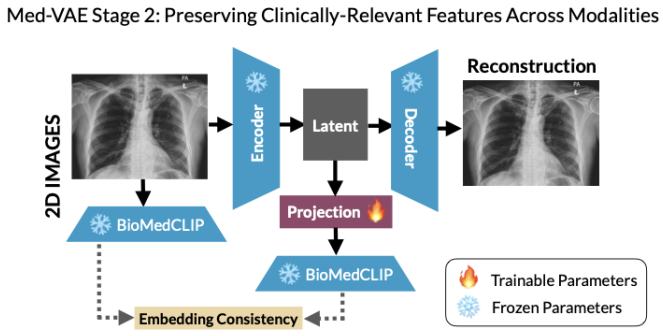


Fig. 8: Original stage 2: latent space alignment on BioMedCLIP embeddings. Reproduced from Varma et al. [1], Fig. 1b.

b) Modified stage 2 — JEPA adaptation.: We replace the BioMedCLIP alignment with JEPA training directly on coronary angiography images (Fig. 9). The architecture has three components, all operating on the encoder's penultimate feature maps (256 channels at 64×64, just before the projection to the single-channel latent):

- **Context encoder**: the stage-1 encoder (frozen decoder), receiving the masked image (40% of patches zeroed) and producing the context features z_c.
- **Target encoder**: EMA copy of the context encoder with `stop-gradient`, encoding the full image. Weights updated as $\theta_t \leftarrow m\,\theta_t + (1-m)\,\theta_c$ ($m = 0.996$).
- **JEPA predictor**: lightweight convolutional network (depth 3, hidden width 512) taking z_c concatenated with a binary mask of target patches, predicting $\hat{z}_t$.

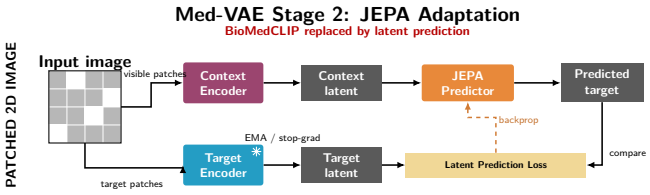


Fig. 9: Modified stage 2: JEPA adaptation — the context encoder predicts the target latent of the EMA encoder (stop-grad).

c) Loss function.: The stage-2 loss is:

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{SIGReg}}\,\mathcal{L}_{\text{SIGReg}}, \tag{2}$$

where $\mathcal{L}_{\text{pred}}$ is the Smooth-L_1 loss between $\hat{z}_t$ and the channel-wise normalised target latent $\tilde{z}_t$, restricted to masked patches $\mathcal{M}$:

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \text{SmoothL}_1\left(\hat{z}_t^{(p)},\ \tilde{z}_t^{(p)}\right), \tag{3}$$

and $\mathcal{L}_{\text{SIGReg}}$ is the SIGReg regularisation [18] on z_c, based on the Epps–Pulley statistic applied to random projections of the embeddings:

$$\mathcal{L}_{\text{SIGReg}} = \mathbb{E}_{\mathbf{v}}\left[T_{\text{EP}}\left(\{z_c^\top \mathbf{v}\}\right)\right]. \tag{4}$$

This term prevents representational collapse without negative pairs. In practice $\lambda_{\text{SIGReg}} = 0.01$.

d) Collapse risks.: JEPA training is exposed to three degenerate modes. **Representational collapse** occurs when the encoder produces a constant representation that the predictor learns to imitate — normalising $\tilde{z}_t$ channel-wise makes this solution explicitly penalised. **Dimensional collapse** corresponds to the progressive extinction of certain latent dimensions; $\mathcal{L}_{\text{SIGReg}}$ detects and corrects this by maintaining an isotropic distribution over random projections. Finally, a **too-low EMA momentum** would cause the target encoder to converge too quickly towards the context encoder, cancelling the learning signal — hence the choice of $m = 0.996$, which keeps the target encoder as a slow-moving average.

At the end of stage 2, one expects a latent space where patches that are anatomically similar produce nearby representations regardless of their spatial position. The predictor acts as a local world model: inferring a masked patch requires understanding the global structure of the image, which forces the encoder to acquire semantically informative representations rather than encoding low-level intensity patterns.

C. Results

Note on this revised version. In an earlier version of this study, the ARCADE test images were part of the pre-training data, and the latent-space analysis (effective rank, number of principal components, k-NN and linear probe)

was computed on the output of the channel-projection head $\texttt{channel_proj}(\texttt{channel_ds}(z) + z)$, a layer trained neither by the stage-1 loss nor by the JEPA loss; its conclusion that stage 2 concentrates the variance on 25 dimensions described a random projection of the latent. Both stages were retrained without the test images, and the representation is now evaluated on the layer that JEPA actually trains, on the 300 test images. The pre-training pipeline was then corrected once more (the augmentation noise reduced to its intended level, random augmentations made to differ across data-loading workers and epochs, fixed masks for the stage-2 validation that selects the checkpoint), and each stage was pre-trained with three seeds: the figures below are means over these three pre-trainings.

a) Reconstruction quality.: During stage 2 the decoder remains frozen, so any displacement of the encoder into a subspace incompatible with the decoder directly degrades reconstruction (Table VI, Fig. 10). Stage 1 reconstructs at 31.1 dB, below the released MedVAE (34.8 dB), which compresses half as much per side and was trained on a million images; stage 2 collapses to 18.6 dB (between 17.6 and 20.5 dB depending on the pre-training), the frozen decoder no longer reading the representation of the adapted encoder.

b) What the encoder has learnt.: We measure how linearly the trained features separate vessels from background: a linear probe (per-channel standardisation, 1×1 convolution, bilinear upsampling to 512×512) is trained on the 1 000 training images, selected on the 200 validation images and evaluated on the test set, with three seeds. The probe must be trained to convergence (with 20 epochs, the MedVAE and stage-1 probes were still improving at the last epoch): we train it for 100 epochs and choose its learning rate, 10^{-3} or 10^{-2}, on the validation Dice, which selects 10^{-2} for all three encoders. Stages 1 and 2 are compared pre-training by pre-training, with a 95 % confidence interval over the three pre-training seeds (Student); MedVAE is a single released model. A probe predicting the 25 artery segments, or using the single-channel latent, reaches a Dice of about 0.03 for every model: a per-pixel linear classifier can neither name a segment nor separate vessels from a single channel. We also measure the dimensionality of the trained features (effective rank of the per-pixel feature vectors and number of principal components explaining 90 % of their variance).

	MedVAE (×4, released)	Stage 1 (×8, from scratch)	Stage 2 JEPA
PSNR / vessel PSNR (dB)	34.8 / 35.5	31.1 / 31.7	18.6 / 22.9
Vessel probe, Dice	0.517 ± 0.005	**0.561 ± 0.006**	0.536 ± 0.011
Effective rank (channels)	251 (512)	147 (256)	156 (256)
Components for 90 % variance	61	46	50

TABLE VI: Reconstruction and representation of the three encoders on the 300 ARCADE test images. The probe and the dimensionality are computed on the encoder features before `conv_out`, the layer trained by the JEPA loss. Stages 1 and 2: mean (± standard deviation) over three pre-trainings; MedVAE: over three probe seeds.

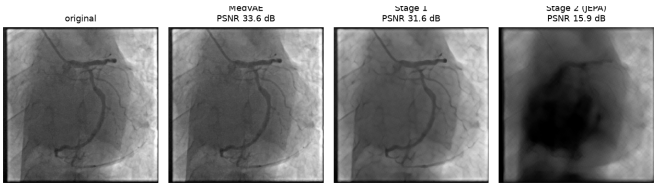


Fig. 10: Reconstruction of a test image by MedVAE, stage 1 and stage 2 (left to right after the original; pre-training seed 42).

JEPA lowers the linear separability of the vessels instead of refining it: relative to the stage-1 encoder it starts from, the probe Dice drops by 0.025 (95 % CI $[-0.045, -0.005]$), for each of the three pre-trainings (-0.019, -0.022, -0.034) and on the validation set as well. The adapted encoder ends up level with the released MedVAE ($+0.019$, CI $[-0.008, +0.045]$). Stage 1 gives the most separable features (0.561, against 0.517 for MedVAE); since the two models differ in architecture, compression factor and training data, we do not attribute this gap to any one of these factors. JEPA does not produce a dimensional contraction either: the effective rank goes from 147 to 156 on average, without a consistent direction across pre-trainings ($+5$, -2, $+26$).

c) Intermediate feature visualisation — RGB PCA.: Projecting the same features on their first three principal components (PC1→R, PC2→G, PC3→B, percentile normalisation [2%, 98%]), following the visualisation protocol of DINO [19] and LeJEPA [18], illustrates these results (Fig. 11). The features of MedVAE and stage 1 follow the vascular tree, stage 1 with a cleaner separation from the background; in those of stage 2 the vessels are still visible, but large regions of homogeneous colour dominate, which is consistent with a coarser description of the image, less suited to the fine local structure that segmentation needs.

d) Summary.: Replacing BioMedCLIP by a self-supervised JEPA objective on ARCADE, with a frozen decoder, degrades both the reconstruction and the linear separability of the vessels relative to the stage-1 autoencoder it starts from, consistently over three pre-trainings; the adapted encoder is no better than the released MedVAE. In our experiments, the most separable features are those of the stage-1 autoencoder trained on ARCADE, before any JEPA training. Both stages were trained for 15 epochs, as in the original study, and their validation losses were still decreasing at the end (best epoch 14 or 15 for every pre-training): longer training could change the levels reported here.

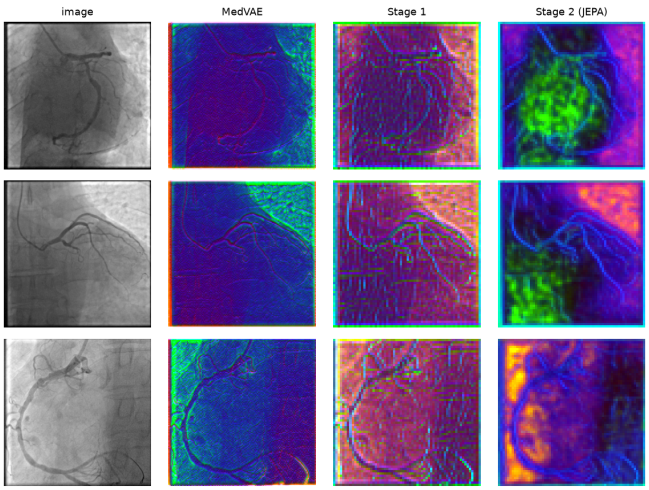


Fig. 11: RGB PCA of the encoder features before `conv_out` for three test angiographies (PCA fitted on 50 other test images; pre-training seed 42). Columns: image, MedVAE, stage 1, stage 2.

V. MedVAE as a Latent Representation for Coronary Segmentation

The high resolution of medical images is necessary to preserve diagnostic details, but it strongly increases the computational and storage cost of deep learning models. An attractive solution consists of compressing images using generic large-scale pre-trained variational autoencoders (VAE) [11], of which MedVAE [1] is a representative example.

This compression remains however intrinsically destructive, and nothing guarantees that it preserves information useful for all downstream tasks. The benefits of MedVAE have been established primarily on classification and detection tasks, which rely on global features. Fine coronary artery segmentation constitutes on the contrary an unfavourable case. Indeed, the structures to be delineated are thin, tubular, and low-contrast, the number of classes is high (26 classes: background and 25 arterial segments), and their distribution is heavily imbalanced. The central question we pose is therefore: **does lossy medical compression preserve sufficient anatomical information to enable multi-class coronary segmentation? A second methodological question arises alongside: at which point in the processing pipeline must the compression be integrated for this information to remain exploitable?**

To answer this, we evaluate on the public ARCADE dataset [2] the extent to which MedVAE can serve as an intermediate representation for segmentation. We compare three strategies accounting for compression against a full-resolution baseline, namely a U-Net [20] trained on original images: (i) segmenting directly from the compressed latent representation with a dedicated segmentation head, (ii) first adapting the compressor to the coronary domain by fine-tuning before segmenting from the latent, and (iii) training a segmentation model on the decompressed images. These three strategies cover both latent-space and reconstructed pixel-space usage, with and without domain adaptation. Our results show that, rendered in pixel space, the compression preserves the information needed for segmentation: segmenting the decompressed images is as accurate as segmenting the originals, whereas a naive downsampling with the same compression rate is not. Segmenting directly from the single-channel latent reaches $96\,\%$ of the baseline's Dice, but the same head fed with a downsampled image of the same size does almost as well: the latent is usable for segmentation, not better than a trivial compression.

Note on this revised version. In an earlier version of this study, segmentation from the latent collapsed to predicting the background everywhere (Dice 0.04), which led us to conclude that the latent was unfit for dense prediction. This collapse came from the training, not from the latent: the head was trained from scratch with a learning rate of 5×10^{-5}, too low to leave the trivial solution, and it had no convolution at the latent resolution, hence almost no spatial context. The earlier version also reported IoU values computed incorrectly (index masks passed to a metric expecting one-hot masks, which yielded IoU values above the Dice), selected models on $20\,\%$ of the training images instead of the official validation split, used a Gaussian-noise augmentation that the current version of Albumentations makes more than ten times stronger than intended, drew the same random augmentations in every data-loading worker and repeated them at every epoch, and relied on a single training run per condition. All results below were obtained after correcting these points, with three training seeds per condition.

A. Methods

a) Data and Preprocessing.: We use the segmentation subset of ARCADE [2] (512×512 greyscale angiographies), whose COCO annotations are converted into dense 26-class masks (background and 25 arterial segments). Models are trained on the $1\,000$ training images, selected on the 200 images of the official validation split, and evaluated on the official test set of 300 images. Intensities are normalised in $[0, 1]$ and training employs standard augmentation (flips, affine transforms, contrast and brightness adjustments, light Gaussian noise of standard deviation 3 to 7 grey levels, CLAHE) via Albumentations [21]. Geometric transforms are applied jointly to the image and its mask to preserve alignment.

b) MedVAE Compressor.: The compressor is the `medvae_4_1_2d` variant of MedVAE [1], which reduces each spatial dimension by a factor of 4: a 512×512 image is encoded into a single-channel 128×128 latent, a factor-16 reduction in the number of values. Unless stated otherwise, its weights are frozen. Conditions B and C use the stage-1 latent (the mean-and-variance bottleneck used for reconstruction), not the output of MedVAE's stage-2 projection layers, which MedVAE intends for downstream tasks.

c) Experimental Conditions.: We compare three strategies accounting for compression against a full-resolution baseline, as well as diagnostic controls (Fig. 12).

A (baseline). A U-Net [20] with a ResNet-34 [22] encoder pre-trained on ImageNet (24.4M parameters), trained end-to-end on original images.

A* (compression alone). The U-Net from condition A, without any retraining, is applied to images first encoded then decoded by MedVAE. This condition isolates the raw effect of pixel-space compression on an unadapted model, independently of any learning.

B (segmentation from the latent). The $128 \times 128 \times 1$ latent produced by the frozen MedVAE encoder feeds a segmentation head trained from scratch: a small U-Net operating at the latent resolution (three downsampling levels, $128 \to 64 \to 32 \to 16$, with skip connections and 64 base channels), followed by two $\times 2$ upsampling blocks back to 512×512 and a 1×1 classifier into 26 classes (7.8M parameters). The convolutions at the latent resolution give the head the spatial context needed to identify a segment, which depends on its position in the coronary tree.

C (domain fine-tuning). Identical to B, but the MedVAE autoencoder is first fine-tuned on ARCADE with a reconstruction loss before being frozen, in order to test whether domain adaptation enriches the latent.

D (segmentation of reconstructed images). A U-Net identical to A is trained not on original images but on their MedVAE-reconstructed versions. Unlike A*, the network here adapts to the compression artefacts.

Trivial-compression controls. To tell what the latent and the reconstruction bring beyond their size, we compress the image trivially at the same rate, by averaging over 4×4 blocks ($128 \times 128 \times 1$, as many values as the latent). **R** trains the head of B, identically, on this downsampled image; it is the control of B and C. **A**$^{\downarrow\uparrow}$ applies A's U-Net, without retraining, to the downsampled image upsampled back to 512×512 (bilinear interpolation); it is the control of A*.

d) Training and Evaluation.: All models minimise a composite loss combining Dice loss and cross-entropy in equal parts:

$$\mathcal{L} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.5\,\mathcal{L}_{\text{CE}}. \tag{5}$$

This choice directly addresses the massive class imbalance. The Dice loss [23] measures the overlap between prediction and ground truth independently of region size, preventing the dominant background from controlling optimisation. Cross-entropy, computed pixel-wise, provides a more stable gradient, especially at the start of training. Optimisation uses AdamW with cosine learning-rate scheduling (initial learning rate 10^{-4} for the U-Nets, 10^{-3} for the heads of B and C, which are trained from scratch), mixed precision, at most 100 epochs and early stopping on the validation Dice. Each condition is trained with three seeds; A* and A$^{\downarrow\uparrow}$ use the three models of A. Differences between conditions are paired by seed and reported with a $95\,\%$ confidence interval over the three seeds (Student). Dice and IoU are computed per class over the whole test set and averaged over the artery segments present in the ground truth or in the prediction (background excluded), which we call the *artery Dice* and *artery IoU*.

B. Results

TABLE VII: Performance on the ARCADE test set (300 images): mean $\pm$ standard deviation over three training seeds (A* and A$^{\downarrow\uparrow}$: the three models of A, evaluation only). The last column includes the background class.

Cond.	Input of the trained model	Artery Dice	Artery IoU	Dice (all)
A	Original images	**0.428 ± 0.008**	**0.309 ± 0.006**	0.450
A*	Reconstructed images (eval. only)	0.424 ± 0.008	0.305 ± 0.006	0.447
B	Pre-trained latent	0.411 ± 0.005	0.292 ± 0.004	0.435
C	Fine-tuned latent	0.410 ± 0.009	0.291 ± 0.008	0.434
D	Reconstructed images	0.427 ± 0.009	**0.309 ± 0.005**	0.450
R	Image downsampled to 128×128	0.405 ± 0.002	0.288 ± 0.003	0.429
A$^{\downarrow\uparrow}$	Downsampled, upsampled (eval. only)	0.389 ± 0.018	0.275 ± 0.014	0.413

Table VII and Fig. 13 summarise test performance; Fig. 14 shows training dynamics and Fig. 16 (appendix) qualitative predictions.

a) Segmentation from the latent works, but no better than from a downsampled image.: With an adequate head and learning rate, conditions B and C reach an artery Dice of 0.411 and 0.410, that is $96\,\%$ of the baseline A (0.428), from a representation sixteen times smaller than the image. They detect the same segments as A (Fig. 13), with the largest gaps on thin and rare branches, and their predictions follow the coronary tree (Fig. 16). The control R shows, however, that this is mostly a property of the 128×128 resolution: the same head, trained on the image average-pooled to 128×128, reaches 0.405, only 0.006 below B (95 % CI $[-0.002, +0.014]$; B is above R for the three seeds on the test set, but not on the validation set). Fine-tuning the compressor on ARCADE has no measurable effect: C $-$ B $= -0.001$ (CI $[-0.036, +0.034]$, with signs differing across seeds).

b) Pixel-space compression is nearly neutral, unlike naive downsampling.: Condition A* measures the effect of compression alone: applying the baseline models, without retraining, to MedVAE-reconstructed images costs only 0.004 of artery Dice (0.424 against 0.428, CI $[-0.009, +0.002]$), and the predictions are nearly identical to those of A (Fig. 16). A trivial compression of the same rate is far more harmful: the same models applied to the image downsampled to 128×128 and upsampled back lose 0.039 (A$^{\downarrow\uparrow}$, CI $[-0.070, -0.008]$), the thin vessels being blurred away. Rendered in pixel space, the $16\times$ compression of MedVAE therefore preserves the anatomical information exploited by the network, which a naive downsampling does not.

c) Adapting to reconstruction brings no measurable gain.: Condition D reaches an artery Dice of 0.427 ± 0.009, against 0.428 ± 0.008 for A: the difference (0.000, CI $[-0.041, +0.040]$, two of the three seeds in favour of A) is within the seed-to-seed variability, and the per-class profiles of A and D are nearly identical. The earlier version of this study reported a gain of D over A concentrated on thin segments; with several seeds and the corrected augmentation this gain disappears. A plausible explanation is the overly strong noise augmentation of the earlier version, which degraded about $30\,\%$ of A's training images while D received them after MedVAE, which removes such noise (Section II).

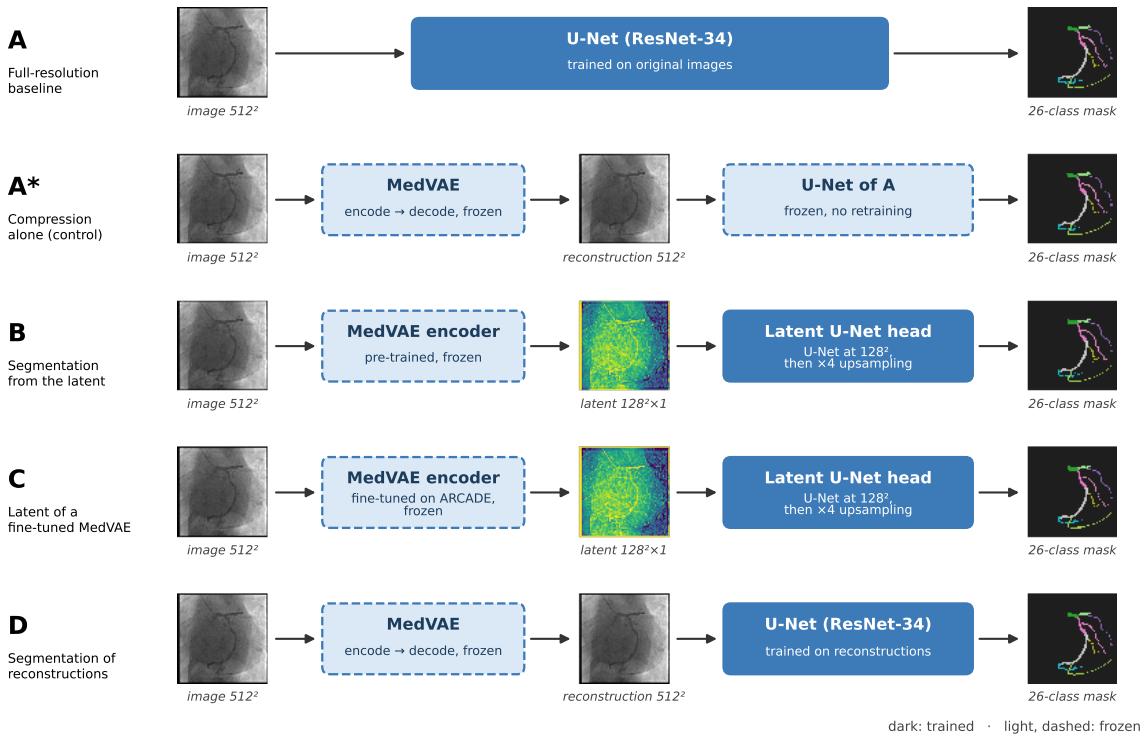


Fig. 12: The five segmentation conditions, illustrated on a test image: the input angiography, the MedVAE latent or reconstruction, and the 26-class mask. Dark boxes are trained, light dashed boxes are frozen. The trivial-compression controls R and $A^{\downarrow\uparrow}$ (not shown) replace MedVAE by a 4×4 average pooling.

d) Segmentation of degraded images.: Section II showed that MedVAE removes Poisson noise; does this make segmentation more robust? We degraded the 300 test images with the same Poisson noise, JPEG compression and blur (levels 0, 3, 6 and 9 of the ten-level grid of Section II) and segmented them with A, A^* and D (three seeds, Fig. 15). MedVAE's denoising does not help: under Poisson noise, A^* is *worse* than A at weak and medium levels (-0.02 to -0.03), the network trained on clean images being more disturbed by the reconstructions of noisy images than by the noise itself, and D behaves like A up to level 6. At the strongest level, Poisson noise and JPEG compression make all three models collapse (artery Dice at most 0.14, D slightly less affected), while the strongest blur roughly halves the Dice (0.18 to 0.21). Robustness depends instead on what the network sees during training: the overly strong noise augmentation of the earlier version made A far more robust (Dice 0.34 under the strongest Poisson noise, against 0.02 with the corrected augmentation), and adding realistic degradations to the training augmentation (Poisson noise, JPEG, blur, applied to $30\,\%$ of the images) restores a large part of it: at the strongest level, $+0.21$ under Poisson noise, $+0.22$ under JPEG compression and $+0.14$ under blur, and $+0.02$ to $+0.08$ at level 6. It costs nothing on clean images: 0.437 ± 0.010 against 0.428 ± 0.008 ($+0.010$, $95\,\%$ CI over seeds $[-0.023, +0.042]$; higher validation Dice for each of the three seeds). Since these degradations are the same families as those evaluated, this last result measures in-distribution robustness.

C. Discussion

Our experiments address both aspects of the problem. Regarding information preservation, a factor-16 lossy medical compression retains sufficient anatomical information for multi-class coronary segmentation when rendered in pixel space: compression alone costs almost nothing ($A^* \approx A$), whereas a naive downsampling of the same rate costs ten times more, and a model trained on reconstructions matches the baseline ($D \approx A$). Regarding the integration point, MedVAE's single-channel latent is directly exploitable for dense prediction, provided the head has convolutions at the latent resolution and is trained with an adequate learning rate: it then reaches $96\,\%$ of the baseline's Dice, with a head three times smaller than the baseline U-Net and inputs sixteen times smaller. The same head reaches almost the same Dice from an image downsampled to 128×128, however: for this task, the latent is a usable input, not a better one than a trivial compression of the same size.

A gap of about 0.02 of Dice remains between the latent and the image, concentrated on thin and rare segments. A multi-channel latent, or intermediate encoder feature maps rather than the sole bottleneck, could reduce it; fine-tuning

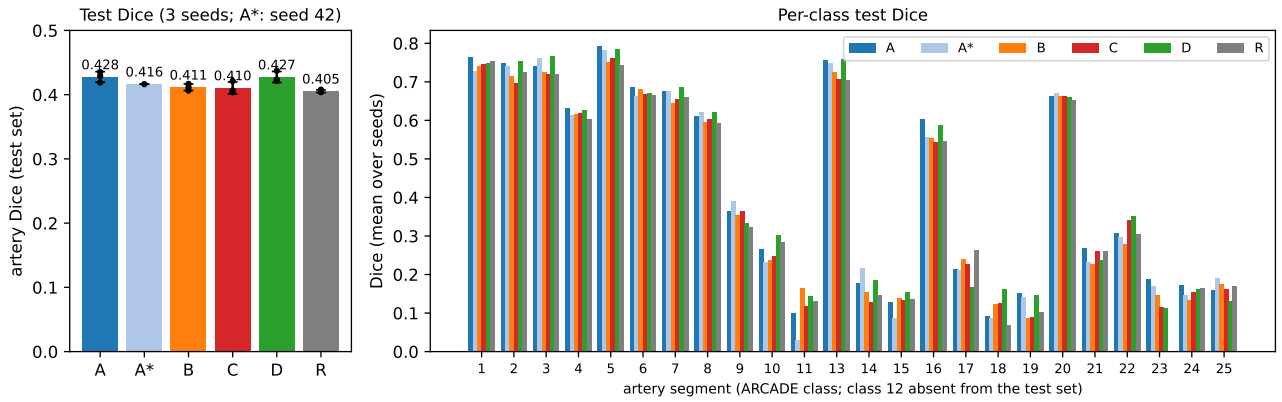


Fig. 13: Left: artery Dice on the test set per condition (bars: mean over seeds, dots: individual seeds). Right: per-class Dice, averaged over seeds. Conditions A, A* and D are nearly indistinguishable; B and C follow the same per-class profile a few points lower, with the largest gaps on thin, rare segments, and the trivial-compression control R is close to them. A*: seed-42 model.

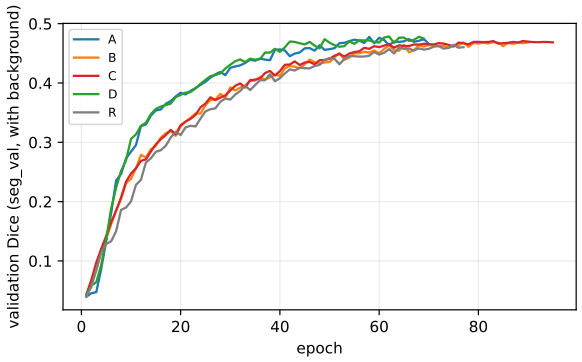


Fig. 14: Validation Dice during training (mean over the three seeds). B and C, which segment from the compressed latent, learn more slowly and plateau a few points below A and D, as does the control R, which segments from the image downsampled to 128×128.

the encoder with a segmentation objective, rather than with the reconstruction objective that had no effect here, is a natural direction. Our results thus delineate the regime in which a generic medical compressor is useful for segmentation: as a pixel-space preprocessor, it is harmless and far more faithful than a naive downsampling; as a latent representation, it is compact and nearly sufficient, but brings no measurable advantage over a downsampled image in our setting.

VI. Perspectives

The experiments presented in this paper suggest a natural research direction: the construction of a *foundation model for radiation-based medical imaging*, unifying the two complementary axes explored here — semantic structuring of the latent space via JEPA and understanding of acquisition-specific degradations.

a) A hybrid JEPA–reconstruction objective.: With a frozen decoder, the JEPA adaptation of Section IV degrades both the reconstruction and the linear separability of vessels. A patch-prediction objective may nonetheless become useful for inducing semantically structured representations in medical imaging, provided it is combined with a reconstruction branch from the outset. Such a hybrid objective would simultaneously predict masked latent patches and reconstruct the input image, keeping the encoder anchored to a space compatible with the decoder. These two objectives are complementary — the first pushes the model to capture high-level contextual relationships between patches, while the second prevents the latent from drifting into a subspace that is no longer decodable. Balancing them via a learnable weight or an annealing schedule would allow progressive control of the semantic/reconstruction trade-off throughout training.

b) Degradations modelled for radiation-based imaging.: The robustness analysis of Section II reveals that the nature of the degradation, and not merely its intensity, determines the model's behaviour. A foundation model should therefore be pre-trained on a degradation distribution that faithfully reflects clinical acquisition conditions. For X-ray and fluoroscopy, we envision a composite forward model grounded in the physical image formation process, combining several sources of variability. Quantum shot noise, dominant at low dose, would be simulated by controlling the photon flux. Luminosity and contrast variations, caused by changes in exposure or detector response, would be modelled by a global brightness factor and a non-linear contrast transform applied to the grey-level dynamic range, both of which have direct physical interpretations in terms of dose and detector gain. Spatially smooth multiplicative and additive fields would account for gain non-uniformities, vignetting, and scattered radiation inherent to X-ray beam heterogeneity. Finally, anisotropic motion blur kernels with orientations drawn from distributions fitted to cardiac and respiratory dynamics would extend the isotropic

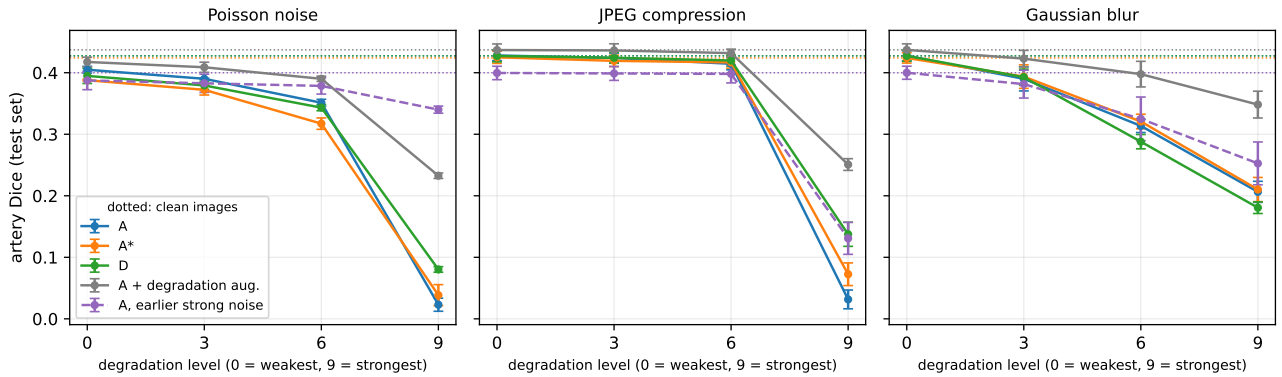


Fig. 15: Artery Dice on degraded test images (mean ± standard deviation over three seeds; dotted lines: clean images). A, A* and D collapse under the strongest Poisson noise and JPEG compression; training A with realistic degradations restores a large part of the robustness that the overly strong noise augmentation of the earlier version (dashed) provided by accident.

blur studied here to clinically realistic directional blurring. Sampling these parameters independently and composing the resulting transforms would yield a rich, physically grounded augmentation space, exposing the model to the full spectrum of artefacts that currently cause reconstruction to collapse, and allowing the JEPA objective to learn representations that are robust to the acquisition variability inherent in radiation-based imaging.

c) Conditional training via a CVAE.: Beyond robustness, an important axis consists of making the model *modality-aware* and *quality-aware*. Section III shows that a quality score computed from the image itself brings no information to an autoencoder of that image; conditioning is only promising with signals that are not a function of the image. The latent space could thus be structured by conditioning the encoder on metadata available in DICOM headers: acquisition modality (fluoroscopy, CT, MRI), patient dose, tube current, or reconstruction parameters. The natural framework for this is a *conditional VAE* (CVAE) [24]: the conditioning signal, concatenated or injected via FiLM layers [10], would guide the encoder to project the image into a sub-region of the latent space corresponding to the acquisition context. Such a model would learn disentangled representations where the anatomical content is separated from the acquisition quality, enabling downstream tasks such as cross-modality transfer, quality-aware retrieval, or targeted denoising adapted to the estimated degradation level. The modality and quality metadata, always available in clinical practice, could thus be converted from a source of latent variability into a structuring prior for training.

d) Downstream validation.: Ultimately, the value of such a foundation model can only be established by evaluating the quality of its representations on concrete downstream tasks. We would aim to validate it across a range of benchmarks spanning classification, detection, and segmentation tasks over multiple modalities. Coronary artery segmentation on the ARCADE dataset constitutes a particularly challenging and clinically relevant test bed: as shown in this paper, it requires fine spatial resolution, tolerance to acquisition artefacts, and discriminative power over a large number of anatomically thin and imbalanced classes — precisely the properties that the proposed training objectives are designed to foster.

VII. Conclusion

Across four complementary axes, we probed how a generic large-scale medical compressor, MedVAE, behaves on the demanding domain of coronary angiography. Measuring restoration against the clean image, our robustness analysis showed that MedVAE reduces Poisson noise, though 3.4 to 6.3 dB less than a simple 5×5 filter, which also keeps vessels, thin ones included, more visible, and restores neither JPEG artefacts nor blur; this noise reduction does not make the segmentation of degraded images more robust. Conditioning the autoencoder on an image-quality score through FiLM, trained and evaluated rigorously, neither improves nor degrades its reconstruction, and the network barely uses the score, which is a function of the image it already sees; fine-tuning MedVAE on ARCADE, in contrast, gains more than 7 dB. Replacing MedVAE's stage-2 BioMedCLIP alignment with a self-supervised JEPA objective, with a frozen decoder, degrades both the reconstruction and the linear separability of vessels relative to the stage-1 autoencoder it starts from, without the dimensional contraction reported earlier. Finally, rendered in pixel space, a $16\times$ lossy compression preserves the anatomical information needed for fine multi-class coronary segmentation: segmenting reconstructed images matches the original ones, unlike a naive downsampling of the same rate; the single-channel latent, with a head that has convolutions at its resolution, reaches $96\,\%$ of the full-image Dice, but no more than a downsampled image of the same size.

Taken together, our results sketch a coherent picture: MedVAE's compression is anatomically faithful and reusable in pixel space, and its latent is usable for segmentation, though not better than a downsampled image; the attempts to make it acquisition-aware or semantically richer did not help in the forms we tested. Several conclusions of the first version of this study were reversed once measurement and training issues

were corrected, which underlines how much such conclusions depend on the protocol: a metric comparing to the right reference, identical training for the compared models, several seeds, and an evaluation on the layer that is actually trained. The way forward we advocate, detailed in our perspectives, is to unify these axes into a single foundation model for radiation-based medical imaging, coupling a hybrid JEPA–reconstruction objective with a physically grounded model of clinical degradations and explicit conditioning on acquisition metadata.

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APPENDIX A
QUALITATIVE PREDICTIONS

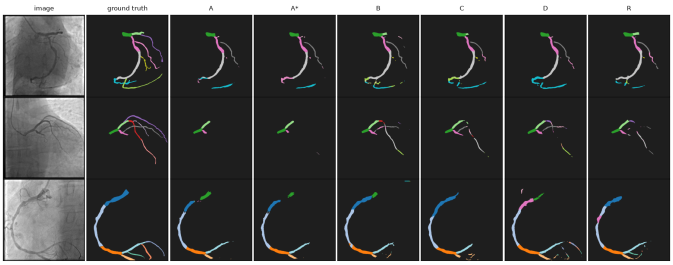


Fig. 16: Predictions of the seed-42 models on three test images (one colour per artery segment). A, A* and D are nearly identical; B and C, which segment from the 128×128 latent, and the control R, which segments from the image downsampled to 128×128, recover the main tree with slightly less regular contours and miss or confuse some thin branches.

APPENDIX B
VESSEL-MASKED PSNR: DETAILS

The masked PSNR (mPSNR) restricts the error computation to vessel pixels only, in order to eliminate the bias of the background, which is largely dominant and clinically irrelevant. ARCADE's polygonal annotations are rasterised into a binary mask (`cv2.fillPoly`), then resized by nearest-neighbour interpolation to preserve their binary nature. The mean squared error is evaluated only over this set of pixels, denoted $\mathcal{M}$:

$$\mathrm{mPSNR}(x, y) = 10 \log_{10} \frac{D^2}{\frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} \left(x_p - y_p\right)^2}, \qquad (6)$$

where D is the intensity range ($D = 1$ for images normalised in $[0, 1]$).

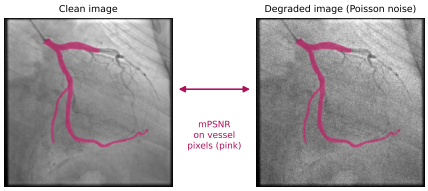


Fig. 17: Principle of vessel-masked PSNR on a real ARCADE example. The clean image and its degraded version share the same vessel mask $\mathcal{M}$ (pink). The error is evaluated only on these pixels ($\approx 2\%$ of the area in this example, 3.5% on average).

APPENDIX C
VISUALISATION OF SYNTHETIC DEGRADATIONS

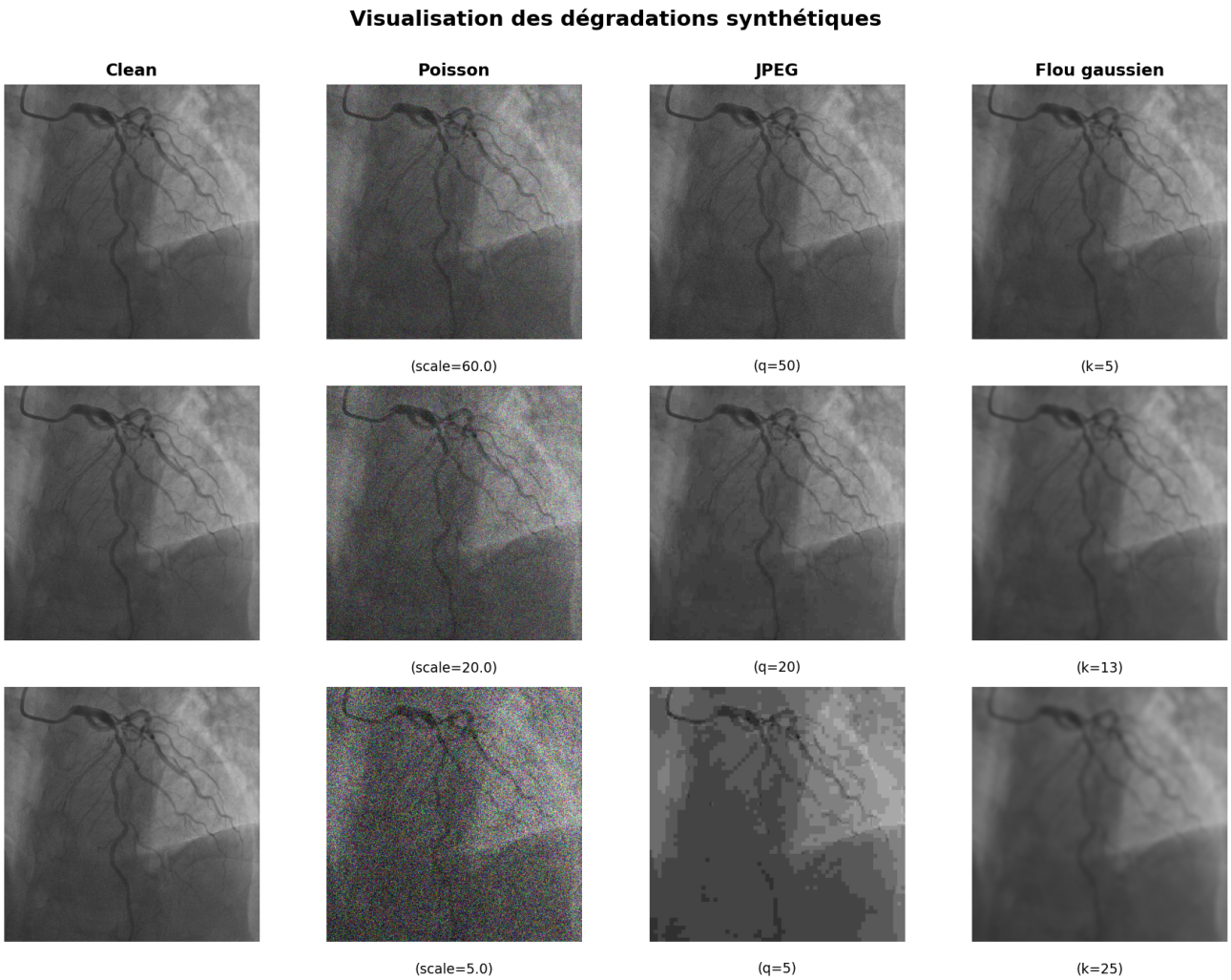


Fig. 18: Visualisation of synthetic input degradations (detail). From left to right: clean image, Poisson noise, JPEG compression, Gaussian blur. Degradation intensity increases from top to bottom, revealing the injection of high frequencies (noise, 8×8 blocks) or conversely their suppression (low-pass smoothing).